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Collaborative Robotic Arms

Digital-Twin Enabled

Graduation Project Thesis

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Abstract

This thesis presents a Digital-Twin-enabled collaborative robotic assembly system integrating an ABB industrial manipulator and an AR4 robotic arm to execute efficient brick assembly through coordinated perception, grasping, handover, and placement. The proposed system combines global scene understanding from an Intel RealSense camera with local “eye-in-hand” sensing using an ESP32-CAM mounted on the AR4, enabling both coarse positioning and fine alignment during grasping via Image-Based Visual Servoing (IBVS). A structured state machine governs end-to-end task allocation and execution, selecting between direct assembly and cross-side handover based on brick and robot allocation logic derived from GUI/database input and global perception.

To achieve accurate interaction in a shared workspace, calibration is treated as a core requirement. AR4 calibration is performed by collecting data at multiple X–Y locations at two height levels (0 mm and 50 mm) and fitting mapping functions using multiple approaches (Radial Basis Function (RBF), linear, and polynomial). Polynomial regression demonstrated superior performance and was validated using training/testing splits. Similarly, global camera calibration based on polynomial regression is applied using samples across varying X, Y, and Z coordinates to enable reliable estimation of brick world pose across the entire table. Perception is implemented through YOLOv8 segmentation for brick and feature detection, while grasp selection is addressed using a transformer-based encoder–decoder trained with both successful grasps (True Grasps) and failure/avoidance cases (False Grasps), enabling negative reinforcement to reduce unstable grasp predictions. During fine control, homography is used to map feature points between the RealSense global view and the ESP32-CAM local view, simplifying tracking and improving approach robustness.

Experimental results demonstrate improved positional accuracy using polynomial calibration, effective brick detection performance using segmentation, increased grasp stability using the True/False labeling strategy, and robust end-to-end execution under a state-machine framework. The thesis concludes with limitations and future directions, including improved domain adaptation, force-aware handovers, and enhanced Digital Twin synchronization for higher fidelity planning and monitoring.

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Chapter 1

Introduction

1.1 Background and Motivation

Collaborative robotic systems are increasingly adopted in modern automation to improve throughput, flexibility, and safety in assembly workflows. While a single industrial robot can achieve high repeatability, multi-robot collaboration enables task decomposition, parallelism, and improved workspace coverage. However, collaboration introduces major technical challenges: accurate calibration between sensing and actuation, reliable perception under real-world conditions, robust grasp planning, and stable closed-loop control for fine alignment. Additionally, system-level logic is required to coordinate roles, manage handovers, and handle uncertainty during execution.

This project addresses these challenges by enabling robust collaboration between an ABB robot and an AR4 robot for brick assembly. The system is Digital-Twin-enabled to support visualization, validation, monitoring, and structured integration between perception, planning, and execution.

1.2 Problem Statement

The problem addressed in this thesis is the design and implementation of a two-robot collaborative assembly system that can:

- Detect and localize bricks in a shared workspace using global perception.
- Select stable grasps and avoid unstable configurations using learned models.
- Execute safe and accurate grasping using hybrid control (point control + visual servoing).
- Perform robust handover from AR4 to ABB when bricks must cross workspace sides.
- Maintain consistent world pose estimation across the table via calibration.
- Coordinate the full workflow using a state machine with clear transitions and recovery logic.

1.3 Project Objectives

The main objectives of the project are:

1. Develop a Digital-Twin-enabled architecture integrating perception, decision logic, and robot execution.
2. Achieve accurate AR4 position correction using data-driven calibration (tested: RBF, linear, polynomial; best: polynomial).
3. Calibrate the global camera-to-world mapping such that brick pose can be computed reliably across workspace positions.
4. Implement robust object detection using YOLOv8 segmentation for bricks and relevant features.

5. Implement grasp detection using a transformer-based encoder-decoder trained with True Grasps and False Grasps.
6. Implement fine alignment for AR4 grasping using IBVS aided by homography mapping between global and local camera views.
7. Implement a state machine that governs setup, allocation, grasping, handover, and assembly reliably.

1.4 Project Scope

This thesis focuses on perception-to-action integration for collaborative assembly using two robotic arms and two camera systems (global and eye-in-hand). The scope includes calibration, detection, grasp selection, visual servoing, handover logic, and system integration under a structured execution framework.

Mechanical design details, low-level ABB proprietary controller programming, and safety certification processes are considered outside the scope, except where required to ensure safe experimental operation.

1.5 Thesis Organization

The thesis is organized as follows:

- **Chapter 2** Reviews key literature in collaborative robotics, handover, calibration, visual servoing, deep-learning-based perception, and Digital Twin integration.
- **Chapter 3** Details the system architecture and methodology, including calibration, detection, grasping, visual servoing, and state-machine logic.
- **Chapter 4** Presents implementation details and experimental results for calibration accuracy, perception performance, grasp prediction quality, and workflow reliability.
- **Chapter 5** Concludes the thesis and proposes future enhancements.

Chapter 2

Literature Review

2.1 Collaborative Robotics and Multi-Robot Assembly

This section reviews foundational concepts in collaborative robots, multi-robot coordination, and task allocation strategies relevant to assembly workflows. Topics include workspace partitioning, coordination architectures, and performance factors such as cycle time, robustness, and fault recovery.

2.2 Robot-to-Robot Handover

Robot handover is a critical capability in collaborative manipulation. The literature emphasizes accurate pose estimation of the held object, coordinated motion planning, and timing constraints to avoid collisions or drops.

2.3 Calibration in Robotic Systems

Calibration links sensed coordinates to robot motion. This work relates to both robot position correction (data-driven mapping) and camera-to-world calibration for pose estimation.

2.4 Deep Learning for object detection and segmentation in robotics

Deep perception methods enable robust detection under varying conditions. Segmentation is particularly useful when precise object boundaries and orientation cues are required for manipulation.

2.5 Grasp Detection and Learning

Modern grasp detection extends beyond classical heuristics by learning grasp quality from data.

2.6 Visual Servoing (IBVS) and Homography-Based Mapping

Visual servoing provides closed-loop correction using image features. IBVS is attractive due to reduced dependence on full 3D reconstruction, while homography enables mapping between camera views when planar assumptions are valid.

Chapter 3

System Design and Methodology

3.1 System Overview and Architecture

The proposed system integrates two robotic arms (ABB IRB 120 and ANNIN AR4) operating in a shared workspace to perform brick assembly tasks. A global Intel RealSense camera provides overall workspace perception, while an AR4-mounted ESP32-CAM mounted as an eye-in-hand camera to enable local visual feedback during final approach and grasping. A digital twin mirrors the operational state for monitoring and validation, while a state machine coordinates the full workflow from allocation to handover and assembly.

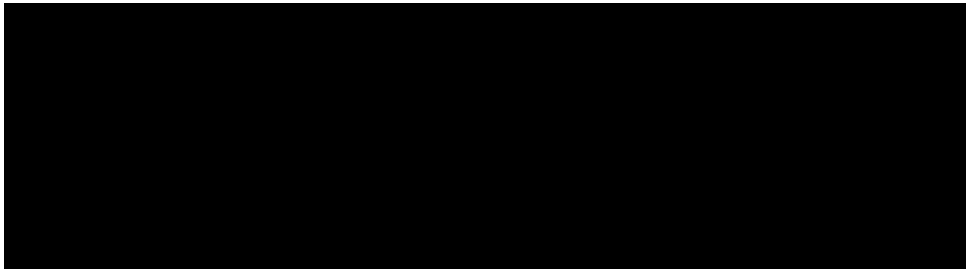


Figure 3.1: System Architecture Diagram showing the integration of ABB, AR4, and perception modules.

3.2 Hardware Components

3.3 Software Architecture

3.4 AR4 Robot Calibration (Data-Driven Correction)

To correct AR4 positioning errors, calibration data was collected at multiple X-Y coordinates for two height levels (0 mm and 50 mm).

3.4.1 Tested Mapping Approaches

Three mapping approaches were evaluated:

1. Linear relations (baseline).
2. Radial Basis Function (RBF) regression.
3. Polynomial regression (Selected).

3.4.2 Polynomial Regression Method

Polynomial regression yielded the best performance. The mapping function estimates corrected pose components as a polynomial function of measured or commanded inputs.

$$\text{Corrected_X} = f_{poly}(X_{cmd}, Y_{cmd}, Z_{level}) \quad (3.1)$$

3.4.3 Validation Metrics

Performance was evaluated using:

- Root Mean Square Error (RMSE) for each axis.
- Maximum absolute error.
- Test-set generalization error.
- Residual plots to identify systematic bias.

3.5 Visual Servoing (Fine Control)

3.5.1 IBVS Control Law

In IBVS, the control command is computed from the image feature error $e = s - s^*$. A standard formulation uses an interaction matrix $L(s)$:

$$v = -\lambda \cdot L^+ \cdot e \quad (3.2)$$

where v is the commanded camera/end-effector velocity, λ is a gain, and L^+ is the pseudo-inverse of the interaction matrix.

Chapter 4

Implementation and Results

4.1 AR4 Calibration Results

Polynomial regression produced the best calibration results among the tested RBF, linear, and polynomial approaches.

Table 4.1: Calibration Error Metrics

Metric	Training Set	Testing Set
RMSE X	[Value] mm	[Value] mm
RMSE Y	[Value] mm	[Value] mm
Max Error	-	[Value] mm

4.2 Discussion

Polynomial calibration improves repeatability and reduces systematic bias versus simpler mappings. True/False grasp labeling improves stability by training the model to reject risky grasps.

Chapter 5

Conclusion and Future Work

5.1 Conclusion

This thesis presented a Digital-Twin-enabled collaborative robotic assembly system. Data-driven calibration using polynomial regression improved accuracy. YOLOv8 and transformer-based grasp models enabled reliable detection and grasping.

5.2 Future Work

1. Force-aware grasping and handover.
2. 6-DoF pose refinement.
3. Formal verification of the state machine.

Bibliography

[1] Author Name, *Title of the Paper*, Journal Name, Year.